

AUSTINE JAISON MANGAN

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PROFESSIONAL SUMMARY

Mechanical Engineering graduate (MACE 2026) with hands-on foundations in SolidWorks, ANSYS FEA, and product development workflows, built through SAE BAJA Collegiate Design Series and an internship at HAL Engine and AERDC, Bengaluru. Demonstrated ability to apply engineering tools under real-world constraints across design, simulation, fabrication coordination, and large-scale operations. Seeking a Graduate Engineer Trainee role to build industry depth across mechanical design and development functions.

TECHNICAL SKILLS

CAD / CAE	SolidWorks, Creo Parametric, Autodesk Fusion 360, ANSYS Workbench (Structural, Modal, Thermal, CFD)
Analysis	FEA (Static, Dynamic/Impact, Modal), Computational Fluid Dynamics (CFD), Multi-Body Dynamics (MBD), Factor of Safety Assessment
Design / Quality	GD&T, Tolerance Stack-up, DFM/DFA, V&V, RCA, CAPA, FMEA Awareness, BOM Management, PLM, AI-Assisted Engineering Workflows.
Tools	MS Office Suite, Technical Drawing & Documentation, Weldment/Assembly Design, Supplier Drawing Review
Operations / Leadership	Cross-Functional Team Leadership, Operations Management, Conflict Resolution, Stakeholder Collaboration, Event Logistics.

INTERNSHIP EXPERIENCE

Hindustan Aeronautics Limited (HAL) – Aero Engine & AERDC Divisions Bengaluru
Engineering Intern July 2025

- Gained primary hands-on exposure to engine overhaul, maintenance, manufacturing, and testing workflows for Kaveri, Adour & HTF3500 programmes at the Aero Engine Division, observing NDT inspection protocols, heat treatment process controls, and in-process quality checks governing a safety-critical regulated production environment.
- Rotated through AERDC (Design & R&D) for the latter half of the internship, gaining exposure to NPI-stage design workflows, tolerance-critical documentation, and aerospace-grade V&V and component qualification systems.

Hindustan Machine Tools Ltd (HMT) – Kalamassery Division Kochi
Manufacturing Intern September 2023

- Gained hands-on exposure to CNC machining operations, DFM principles, machining tolerances, surface finish specifications, and production documentation workflows across an industrial manufacturing facility.

ENGINEERING EXPERIENCE

Chassis Team Lead & Test Driver – H-Baja & M-Baja, SAE MACE Kochi & Indore
SAE Collegiate Design Series (Society of Automotive Engineers) 2024 – 2026

- Engineered a full SAE-compliant ATV tubular space-frame chassis in SolidWorks, applying GD&T and tolerance stack-up principles to meet dimensional and assembly accuracy requirements across all fabricated sub-assemblies and weld joints.
- Reduced overall frame weight by **17%** to a 30 kg final mass, validated against 5G dynamic loads, by executing iterative ANSYS FEA across front impact, side impact, rear impact, rollover and drop load cases, feeding structured RCA back into geometry revisions across multiple prototype cycles.
- Validated structural performance as primary test driver, correlating ANSYS predictions with physical test data; managed supplier drawing reviews, bridging CAD design intent with shop-floor fabrication under competition deadlines.

Team Manager & Design Contributor – Mega ATV Championship, SAE MACE Goa
Mega - ATV Championship 2023 – 2024

- Contributed to component CAD modelling, structural analysis, and fabrication drawings across the full NPI cycle; managed project budget and procurement logistics; co-drove during dynamic events to support V&V data collection.

KEY PROJECT

All-Terrain Multi-Modal Morphobot *Final Year Project, MACE*

- Verified structural integrity of 3D-printed PLA load-bearing components in ANSYS, achieving FOS of **2.375** under 3G hard-landing loads, and reduced parasitic drag by **9%** through iterative CFD optimisation of wheel-spoke geometry in SolidWorks Flow Simulation.
- Validated autonomous mode transitions within a 2.5 s acceptance window by correlating Multi-Body Dynamics (MBD) simulation outputs with prototype test data, closing the simulation-to-hardware loop.

LEADERSHIP & PROFESSIONAL ACTIVITIES

- **Vice Chairman**, SAE MACE (2025–Present) – Strategic planning and coordination across 6+ concurrent engineering project teams and national-level automotive competitions; liaison between student teams, faculty advisors, and SAE India.
- **Director of Operations**, MACE MUN XII (2025) – Delivered a 200+ delegate international conference **15% under budget**; managed 50+ volunteers across 8 departments with zero operational escalations.
- **Secretary**, NSS Unit 117, MACE (2024–25) – Mobilised and directed a 400-member volunteer body; coordinated large-scale multi-agency residential programmes.

EDUCATION

Mar Athanasius College of Engineering (MACE) Kothamangalam, Kerala
B.Tech in Mechanical Engineering 2026

- Best Outgoing Student, Mechanical Dept. (2026) | Departmental Seminar Topper (2026)
- 4th Place – CAM Competition, SAE National Convention 2024, Chennai | 3rd Place – Engineering Drawing, SAEISS Tier II 2023